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CLASS : SY-B (67)

Q1. What is a Data Science Pipeline? Explain its key stages with examples. Answer:

A Data Science Pipeline is a systematic process that involves a sequence of steps used to collect, process, analyze, and build models from data. It helps in converting raw data into meaningful insights and predictions.

Key Stages of Data Science Pipeline: Data Collection: In this stage, data is gathered from various sources such as databases, CSV files, APIs, or sensors. Example: Collecting student marks data from a CSV file. Data Cleaning (Preprocessing): Raw data may contain missing values, duplicate records, or incorrect data. These issues are handled in this stage. Example: Filling missing values or removing duplicate rows. Exploratory Data Analysis (EDA): Data is analyzed using statistical methods and visualization tools to understand patterns and relationships. Example: Plotting graphs to understand distribution of marks. Feature Engineering: New features are created from existing data to improve model performance. Example: Creating “Total Marks” from individual subject marks. Feature Scaling: Data is scaled to bring all values into a similar range. Example: Normalizing salary values. Model Building: Machine learning algorithms are applied to train models. Example: Using Linear Regression to predict salary. Model Evaluation: The model’s performance is tested using metrics like accuracy or error rate. Example: Checking prediction accuracy on test data.

Q2. What is Exploratory Data Analysis (EDA)? How do univariate and bivariate analyses help in understanding the dataset? Answer:

Exploratory Data Analysis (EDA) is the process of analyzing and summarizing datasets using statistical and graphical methods. It helps in understanding the structure, patterns, and relationships present in the data.

Types of Analysis: Univariate Analysis: It involves analyzing a single variable at a time. It helps in understanding the distribution, central tendency, and spread of the data. Example: Finding the average age of students or plotting a histogram of marks. Bivariate Analysis: It involves analyzing two variables together to understand their relationship. Example: Studying the relationship between age and salary using a scatter plot. Importance of EDA: Helps in identifying patterns and trends Detects outliers and anomalies Improves decision-making for model building

Q3. Differentiate between Label Encoding and One-Hot Encoding. When should each be used? Answer:

Label Encoding and One-Hot Encoding are techniques used to convert categorical data into numerical form.

Differences: Basis Label Encoding One-Hot Encoding Definition Assigns a unique number to each category Creates separate binary columns for each category Example Red=0, Blue=1, Green=2 Red=[1,0,0], Blue=[0,1,0] Data Type Single column Multiple columns Order Implies order (may be misleading) No order is implied Usage: Label Encoding: Used when categories have a natural order (e.g., Low, Medium, High). One-Hot Encoding: Used when categories do not have any order (e.g., Colors, Cities).

Q4. Why is feature scaling required? Compare normalization and standardization. Answer:

Feature scaling is required to bring all numerical features to a similar range. It helps machine learning algorithms perform efficiently and improves model accuracy.

Types of Feature Scaling: Normalization: It scales data between 0 and 1.

Example: Scaling age values between 0 and 1.

Standardization: It transforms data so that it has mean = 0 and standard deviation = 1.

Q5. What is feature engineering? Explain how creating new features can improve model performance. Answer:

Feature Engineering is the process of creating new input features from existing data to improve the performance of machine learning models.

Explanation:

Raw data may not always provide the best information for prediction. By creating meaningful features, the model can learn better patterns.

Examples: Combining features: Total Income = Salary + Bonus Extracting features: Date → Day, Month, Year Transforming features: Log transformation of skewed data Benefits: Improves model accuracy Helps in better pattern recognition Reduces complexity of the model

```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder, OneHotEncoder
from sklearn.preprocessing import MinMaxScaler, StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# Step 1: Create Dataset
data = {
    'Age': [20, 25, 30, 35, 40],
    'Salary': [20000, 25000, 30000, 35000, 40000],
    'City': ['Pune', 'Mumbai', 'Delhi', 'Pune', 'Delhi']
}

df = pd.DataFrame(data)
print("Dataset:\n", df)

# Step 2: EDA
print("\nBasic Info:\n", df.info())
print("\nStatistics:\n", df.describe())
```

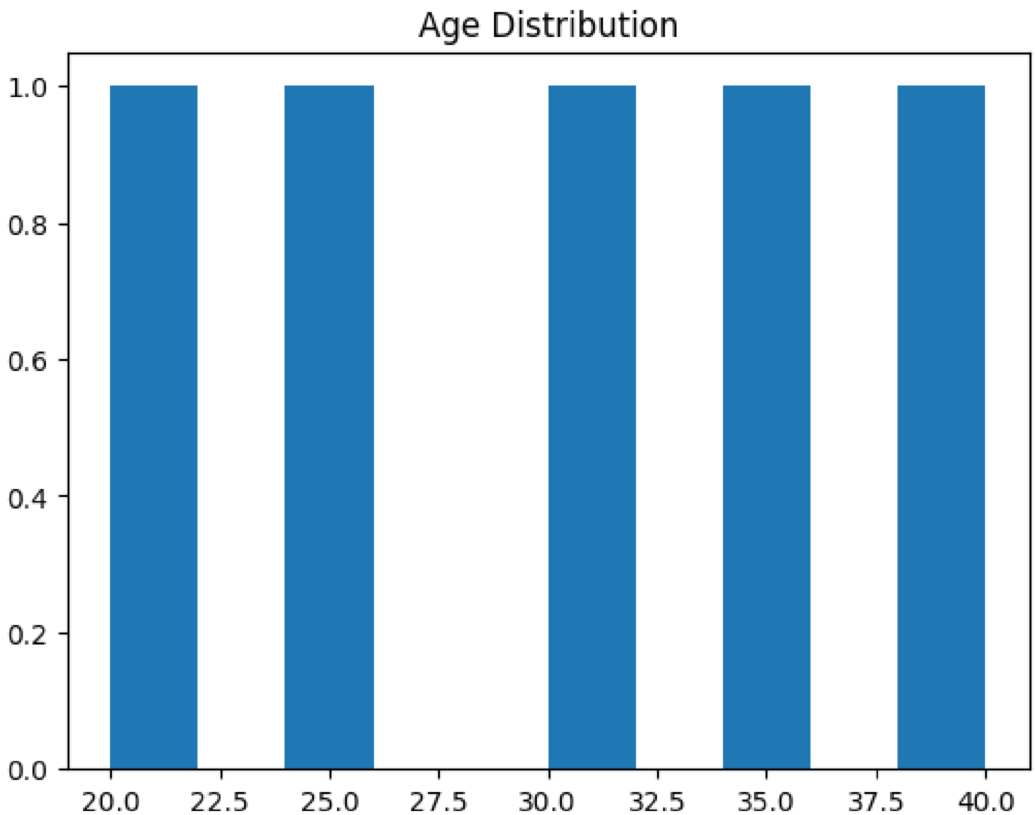
```
Dataset:
    Age  Salary    City
```

```
0   20   20000    Pune
1   25   25000  Mumbai
2   30   30000   Delhi
3   35   35000    Pune
4   40   40000   Delhi
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype 
---  ------  --------------  ----- 
 0   Age     5 non-null      int64 
 1   Salary  5 non-null      int64 
 2   City    5 non-null      object
dtypes: int64(2), object(1)
memory usage: 252.0+ bytes

Basic Info:
 None

Statistics:
              Age       Salary
count   5.000000      5.00000
mean   30.000000  30000.00000
std     7.905694   7905.69415
min    20.000000  20000.00000
25%    25.000000  25000.00000
50%    30.000000  30000.00000
75%    35.000000  35000.00000
max    40.000000  40000.00000

# Plot
plt.hist(df['Age'])
plt.title("Age Distribution")
plt.show()
```




```
# Step 3: Label Encoding
le = LabelEncoder()
df['City_Label'] = le.fit_transform(df['City'])
print("\nLabel Encoded:\n", df)
```

```
Label Encoded:
    Age  Salary    City  City_Label
0   20   20000    Pune           2
1   25   25000  Mumbai           1
2   30   30000   Delhi           0
3   35   35000    Pune           2
4   40   40000   Delhi           0
```

```
# Step 4: One-Hot Encoding
df_onehot = pd.get_dummies(df, columns=['City'])
print("\nOne-Hot Encoded:\n", df_onehot)
```

```
One-Hot Encoded:
    Age  Salary  City_Label  City_Delhi  City_Mumbai  City_Pune
0   20   20000           2       False        False       True
1   25   25000           1       False         True      False
2   30   30000           0        True        False      False
```

```
3   35   35000           2       False         False       True
4   40   40000           0        True         False      False

# Step 5: Feature Scaling
scaler = MinMaxScaler()
df[['Age_scaled']] = scaler.fit_transform(df[['Age']])

std_scaler = StandardScaler()
df[['Salary_scaled']] = std_scaler.fit_transform(df[['Salary']])

print("\nScaled Data:\n", df)


Scaled Data:
    Age  Salary    City  City_Label  Age_scaled  Salary_scaled
0   20   20000    Pune           2        0.00      -1.414214
1   25   25000  Mumbai           1        0.25      -0.707107
2   30   30000   Delhi           0        0.50       0.000000
3   35   35000    Pune           2        0.75       0.707107
4   40   40000   Delhi           0        1.00       1.414214

# Step 6: Model Building
X = df[['Age']]
y = df['Salary']

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2)

model = LinearRegression()
model.fit(X_train, y_train)

LinearRegression()

# Prediction
pred = model.predict(X_test)
print("\nPredictions:", pred)


Predictions: [20000.]
```